Session-II: Knowing the Hydrological Unit (HU) – July 2nd week

Program Schedule

S.No	Time	Content	Method	Materials	Expected outcome
1	15 min.	Recap.	Large group discussion	Charts, markers and clips.	Participants recollect the contents of the first session.
2	10 min.	Sharing BBE results.	Large group	Charts and markers	Participants know their awareness levels as a group on various FWS topics.
3	90 min.	HESA- HU walk	Field visit, and small group discussion	Visuals, charts and markers.	Discuss rock formations and its extent.
4	25 min.	Water cycle of the Hydrological Unit	Small group discussion	Visuals, charts and markers.	Participants discuss water cycle and its significance.
5	15 min.	Group dynamics- Leadership qualities.	Demonstration	Glass tumblers, pebbles, clay and sugar.	This game helps the participants to understand the need for leaders and the qualities s/he should have..
6	30 min.	Short Study- Seed treatment.	Large group discussion	Charts and markers.	Participants discuss the various seed treatment methods.
7	One week	Seed germination test.	Demonstration	Plant seeds, soil, water, cotton cloth, newspaper, rubber bands, gunny cloth, polythene covers, newspaper, and mug.	Participants compute germination rates.
8	10 min..	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session, participants will be able to:

- Recap contents of the previous session,
- Share Pre-BBE results,
- Walk along the HU and discuss hydrological cycle of the HU,
- Discuss water cycle of the Hydrological Unit, and
- Understand the importance of leadership qualities.

Process

1. Recap Time: 15 min**.**

Objective : Participants recap the contents discussed in the previous session.

Content : Recap session one contents.

- Introduction of the participants and sharing of expectations,
- Pre Ballot Box Exercise,
- FWS objectives,
- Norm setting,
- Group formation,
- Elements of farmer decision making, and
- Importance of people's institutions.

Method : Large group discussion

Materials : First session contents, charts, and markers.

Process

- Welcome participants to the second session.
- Start the session with FWS pledge.

FWS pledge
We, the members of the Farmer Water School take an oath that:
The groundwater resources in our HU collectively belong to us.
Together, we will manage our groundwater resources judiciously.
We will take all the necessary measures to prevent groundwater pollution.
We will attend all the 14 sessions of Farmer Water School without fail.
We will disseminate all the learning from FWS to the larger community.
We preserve groundwater resources for our future generations and we take a pledge in this regard.

- In the large group, encourage each participant to share a content that was discussed in previous session and the learning from that one.
- As you facilitate the recap, ask one of the participants to list the contents on the chart paper.
- Briefly discuss the contents and the learning from the previous session.

2. Sharing BBE results Time: 10 min.

Share the BBE results with the participants. Explain the process of analyzing BBE results

3. HESA – HU walk Time: 90 min.

Objective : Participants walk along the HU and discuss hydrological features of the HU.

Content :

- Hydrological cycle in the HU,
- Rock formations and its extent, and
- Soil formations.

Methods : Field visit, and small group discussion.

Materials : Visuals, charts, and markers.

Process

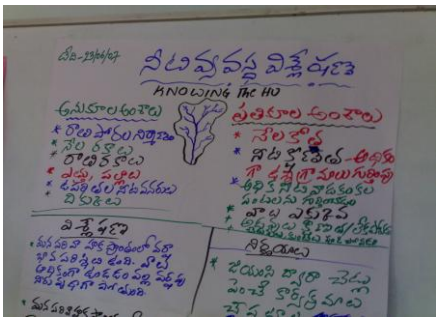
- Inform the participants that they will work in their respective learning groups during this activity.
- Take the farmers to the field (HU area only) and request them to observe and analyze the different rock and soil formations, other factors that influence the hydrological cycle of the HU. Ask the farmers to work in their respective small groups.
- Suggest the farmers to note down their observations during the field walk.
- Encourage farmers to discuss about rock weathering, rock formations and its extent, soil formations, different types of soils, factors influencing groundwater recharge and relevant information. Give 30 minutes for field walk.

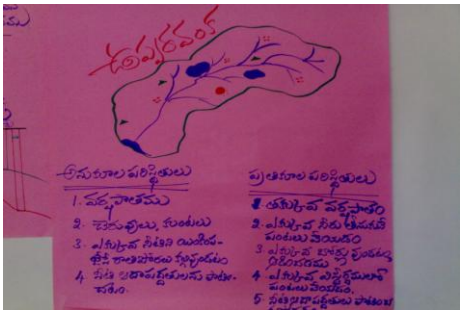
<table>
<tr><td colspan="2"><u>Hydro Eco-System Analysis</u>

Name of the group: Date:

Place/venue:
  </td></tr>
<tr><td>+ <u>Beneficial factors(Recharge)</u>
• List of their observations.</td><td>- <u>Non beneficial(Recharge/Discharge)</u></td></tr>
<tr><td rowspan="2"><u>Analysis</u>:</td><td><u>Decisions</u></td></tr>
<tr><td><u>Action Plan (include role of HUNs/GMCs)</u></td></tr>
</table>

- Later, request farmers to work in small groups and analyze their observations. The above chart could be used to present the analysis.
- Ask the participants to share their observations and analysis in large group. Generate discussion to facilitate a good understanding of the HU.





HESA Frame work

4. Water cycle of the Hydrological Unit Time: 25 min.

Objective : Participants walk along the HU and discuss hydrological features of the HU.

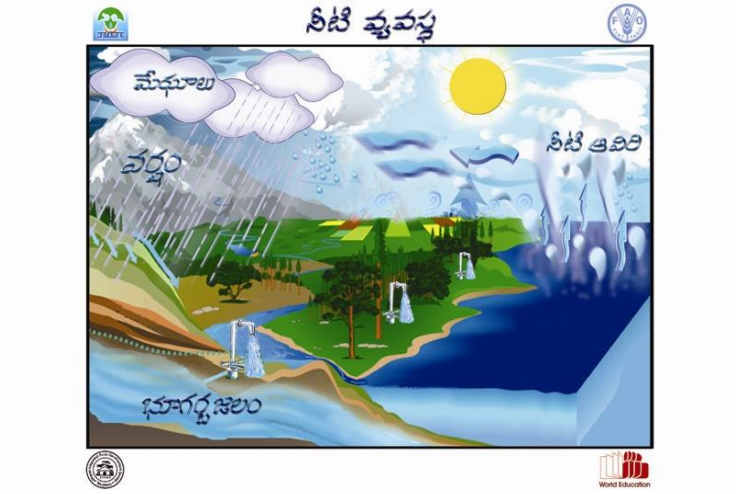
Content : Water cycle of the Hydrological Unit.

Methods : Small group discussion

Materials : Visual, charts, and markers.

Process

- Encourage participants to work in small groups.
- Show the following visual and encourage discussion on how rain occurs and percolation of water into the ground.



- Summarize the discussion by stating that water from oceans and large surface water bodies evaporates due to heat from the sun which again form clouds that condense to fall as rain.

5. Group dynamic- Leadership qualities Time: 15 min.

This game helps participants to identify leaders who:

- Trust and respect others,
- Have a vision, and
- Encourage every one's participation in GMC/HUN.

Game: Three tumblers

- Take four transparent tumblers (glass or plastic) and fill them with water.

- Ask the participants to put pebbles in the first glass, some water in the second, some wet clay in the third, and sugar in the fourth glass.
- Ask them to explain what happened in each case.
- Explain that the pebble represents the autocratic leader, who is dominating and does not mingle with other members.
- Adding water to the second glass represents the dummy leader in a group; there is no difference between the leader and other members in the group.
- The wet clay in the third glass represents the anarchic leader who often promotes chaos and dissidence in the group, making clean water muddy. The group is spoilt just as the water gets muddy.
- The sugar in the fourth glass represents the democratic leader who mingles with the group, improves its quality (sweetens the water), and functions like a genuine people's representative.

After the game, ask participants the following question:

- Who is a good leader in this game and why?

Write the responses on a chart and summarize the discussion with the list of characteristics of a good leader.



6. Short Study: Seed treatment Time: 30 min.

Objective : Participants discuss the various seed treatment methods.
Content : Seed treatment and its usage.
Method : Large group discussion
Materials : Charts and markers.
Process

- Ask participants the following question:
  - What are the ways of controlling pest and diseases at the stage of seed selection i.e. before sowing?"
- List participant's responses on a chart paper and encourage them to discuss about the importance of seed treatment. Share the following information about the importance and benefits of seed treatment.

Tips to facilitator

Information on Seed treatment

- The concept of seed treatment is the use and application of biological and chemical agents that control or contain primary soil and seed borne infestation of insects and diseases that pose devastating consequences to crop production, improving crop safety leading to good establishment of healthy and vigorous plants resulting in better yields.
- Seed treatment refers to the application of fungicide, insecticide, or a combination of both, to seeds so as to disinfect them from seed-borne or soil-borne pathogenic organisms and storage insects. It also refers to the subjecting of seeds to solar energy exposure, and immersion in conditioned water.

- Next, ask participants to discuss in small groups about the advantages of seed treatment.
- Encourage each group to present their group work to large group and summarize the discussion by sharing the following advantages of seed treatment.

Tips to facilitator

Benefits of Seed Treatment

- Prevents spread of plant diseases.
- Improves germination.
- Controls soil insects.
- Ensures uniform seedling emergence.
- Protect seeds or seedlings from early season diseases and insect pests improving crop emergence and its growth.
- Use of plant growth hormones may enhance crop performance during the growing season.
- Rhizobium inoculation enhances the nitrogen fixing capability of legume crops, and their productivity.

- Next, encourage participants to discuss on types of seed treatment. Invite one or two participants who have been practicing seed treatment and encourage them to share their experiences and request him or her to demonstrate the process of seed treatment to larger audience.
- Discuss the seed treatment practices for the major crops in the area.

7. Seed germination test

Time: one week

Introduction :

Seeds are an important and critical input for getting better crop quality and maximum yield. As seed quality is the key factor in plant growth and yield, it is important to know the germination rate by various methods before sowing. Farmers will incur losses in labor cost, seed cost, along with precious time if the seeds are of inferior quality. Therefore, in order to avoid the above inconvenience farmers should be made aware of importance of seed germination and appropriate farmer friendly methods before sowing.

Objectives :

By the end of this exercise, participants will be able to:

- Compute germination rates. and
- Select appropriate farmer friendly germination methods.

Method :

- Large group discussion to introduce the topic.
- Experimentation in small groups.

Materials :

Plant seeds, soil, water, cotton cloth, newspaper, rubber bands, gunny cloth, polythene covers, newspaper, and a mug.

Procedure :

Introduce and discuss the topic in large group. Divide the participants into groups.

Ask the groups to practice the following four methods. They are:

1. Cloth method
2. Tray method
3. Paper method and
4. Petri dish method

Below are a few alternate methods of performing seed germination test. Choose the ones feasible. Around fifty seeds can be used for each germination test and the duration of the test will be for seven days.

Cloth method:

- Tie the seeds loosely for aeration in a wet white cloth.
- Sprinkle water regularly to maintain adequate moisture.

Tray method:

- Take well mixed soil and organic matter in the ratio of 3:1 into a tray.
- Sow the seeds in rows.
- Water the trays regularly to maintain soil moisture.

Paper method:

- Spread the plastic sheet on the ground.
- Place a wet newspaper over the plastic sheet.
- Arrange the seeds on the wet paper.
- Again, cover the seeds with wet news paper.
- Then, roll the sheet. Tie one side with rubber band for moisture retention and other side left open for better aeration.

Petri dish method:

- Spread the circular moist blotter paper in two petri dishes.
- Arrange seeds each in the petri dishes.

- Cover with lids.
- Sprinkle water regularly to maintain moisture.

8. Evaluation Time: 10min.

- Ask the following questions to evaluate the day's learning. Pose the questions to the large group:
    - What did you like about the field visit?
    - What are the factors influencing rainfall in the HU?
    - What are the different elements in the HESA framework?
- Encourage each participant to share one or two points of their day's learning and thank them for their enthusiasm during the HU walk.